

Nivi Jha

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Education

Jaypee University of Information Technology

Bachelor of Technology – Computer Science & Engineering (CGPA: 8.5 / 10)

2023 – 2027

Relevant Coursework: DSA, Machine Learning, OOPS, Operating Systems, DBMS, Cloud Computing, Computer Networks

Work Experience

Summer Intern | IIT Delhi, India

June 2025 – July 2025

- Engineered a Bash and cron daemon collecting CPU, memory, and disk telemetry every 60 seconds, reducing manual infrastructure checks by 80%
- Implemented a Python security agent, achieving **sub-2-second latency** in forwarding structured JSON audit logs for over 100 daily SSH login events.
- Completed **network automation** workflows via Cisco DevNet, consuming REST APIs and containerizing services with Docker for reproducible deployment

Projects

CardioVision | Coronary Artery Stenosis Detection using Deep Learning

Repo | Live

Tech: Python, FastAPI, PyTorch, YOLOv8, OpenCV, Pandas, Grad-CAM, Explainable AI (XAI), Computer Vision

- Trained and benchmarked **3 YOLOv8 variants** (n/s/m) on **1,500 labeled coronary angiography images** from the ARCADE dataset; best model achieved **94.4% mAP@0.5** and **0.91 F1-score**, validated on CADICA benchmark for generalization
- Built a FastAPI inference server wrapping dual YOLOv8 detection + segmentation pipelines, achieving **sub-200ms prediction latency**; served model outputs as structured JSON with stenosis severity grading (5-class) to the frontend
- Authored Python script automating COCO to YOLO annotation conversion, thereby reducing manual labeling prep time by over 7 hours per project dataset

Chat Constellation | Graph-Based Conversational Interaction Visualization

Repo | Live

Tech: React.js, Node.js, Express, MongoDB, OpenAI API, React Flow, JWT, Google OAuth 2.0, Docker

- Engineered a full-stack AI chat app with non-linear graph UX using React Flow, replacing scroll history with real-time sync via Express/MongoDB, supporting 10+ concurrent users.
- Implemented JWT + Google OAuth 2.0 authentication, NoSQL injection prevention via automated query sanitization, and complete account deletion with cascading cleanup of all associated thread and message data

AI Product Review Analyzer | NLP-Based Review Understanding System

Repo

Tech: React.js, AWS Lambda (Node.js), API Gateway, DynamoDB, AWS Comprehend, CloudWatch, IAM

- Designed a **serverless NLP pipeline** on AWS — Lambda + API Gateway + DynamoDB + Comprehend — processing product reviews with **4-class sentiment classification** (Positive/Negative/Neutral/Mixed) at zero idle infrastructure cost
- Architected a decoupled serverless solution with AWS Lambda and DynamoDB, enabling independent scaling

Skills

Programming: Python, Core Java, C, C++, JavaScript
Web & Backend: Node.js, Express.js, React.js, REST APIs
Databases: MongoDB, MySQL, DynamoDB
Systems & Tools: Docker, Linux, Git, GitHub

ML & AI: NumPy, Pandas, PyTorch, FastAPI, YOLOv8, OpenCV, Grad-CAM
Cloud Platform: AWS Lambda, AWS Comprehend

Leadership & Achievements

- Research Publication:** Co-authored research on explainable deep learning for coronary artery stenosis detection using YOLOv8 and Grad-CAM; manuscript currently under editorial review at an SCIE-indexed journal
- Web Dev Team Lead**, JYC Technical Club — Led a team of **5 developers**; mentored **3 junior contributors** on React.js and Git
- LeetCode** — **200+ problems solved** | Contest Rating: **1574** | Top **27.68%** globally
- AWS Academy Graduate – Cloud Foundations** | Amazon Web Services (AWS) | 2025